

you must choose a hub that can handle all different protocols, and will have future room to expand your system without having to replace the controller.

You'll also have to keep in mind what mode of connectivity you would prefer to use with your hub. In case you plan to control all your devices from outside your home more frequently, a Wi-Fi option would make more sense over Bluetooth, even after factoring in higher costs for Wi-Fi connectivity and lower energy consumption for Bluetooth. And in the case you do use a smartphone app instead of a dedicated remote, the availability of the app on your OS' app store is also a deciding factor.

Communication protocols

There are several protocols for wireless communication between the hub and the connected devices, such as IR (infrared used in various remotes), Wi-Fi, Z-Wave and ZigBee. While the earlier systems, like X-10, made use of household electrical wiring, which powers lights and appliances, to send digital data between X10 devices, newer protocols try and provide maximum flexibility by using wireless technology because, apart from the difficulty of installation of wired protocols, they're faster, less intrusive and unsightly, and easier to move.

Smart devices, switches and sensors

Depending on what you want to automate and how you intend to do it, you will need to install smart wall switches and lights, sensors and more complicated smart devices like smart lighting systems, smart thermostats and smoke detectors. In many cases, separate actuators, which are needed for physically moving parts such as moving TV stands, need to be installed.

The simplest actuator is smart switches, such as Belkin's WeMo Insight Switch and Light Switch. These small devices can be plugged into any power outlet and allow you to control the connected appliance wirelessly via an app, through either Wi-Fi or Bluetooth. These smart switches are the perfect introduction to smart-home technology for beginners. No need for programming or expertise with home wiring systems, and you don't even need a separate hub/controller, either. An empty outlet and a smartphone, and you've got everything you need. You'll just need to plug in the smart switch, use the app to configure the switch to connect to the Wi-Fi, and you've got yourself your remotely controllable device.

The list of smart devices on the market is never-ending, too. Smart bulbs are also another cheap and easy way to start with smart-home automation.